FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: transformer60909
Model status: BOUNDED_IEC60909_RADIAL_TRANSFORMER_ENVELOPE
Captured UTC: 2026-10-26T06:04:33.629Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: ratedHighVoltageV | Entered value: 13800
Input path: ratedLowVoltageV | Entered value: 480
Input path: transformer.sourceReference | Entered value: ILLUSTRATIVE: replace with actual approved 2500 kVA transformer OEM nameplate and loss test
Input path: transformer.ratedMVA | Entered value: 2.5
Input path: transformer.shortCircuitVoltagePercent | Entered value: 5.75
Input path: transformer.ratedLoadLossKW | Entered value: 18
Input path: transformer.ratedTapRatio | Entered value: 1
Input path: utility.sourceReference | Entered value: ILLUSTRATIVE: replace with max/min POI equivalent utility study and matching X/R
Input path: utility.maximumEquivalentMVA | Entered value: 350
Input path: utility.maximumXOverR | Entered value: 10
Input path: utility.minimumEquivalentMVA | Entered value: 220
Input path: utility.minimumXOverR | Entered value: 8
Input path: voltageFactors.maximum | Entered value: 1.1
Input path: voltageFactors.minimum | Entered value: 0.95
Input path: feeder.sourceReference | Entered value: ILLUSTRATIVE LV cable sizing and installation; replace with approved conductor R20/X/route/temperature
Input path: feeder.lengthM | Entered value: 25
Input path: feeder.resistanceOhmPerKmAt20C | Entered value: 0.16
Input path: feeder.reactanceOhmPerKm | Entered value: 0.08
Input path: feeder.alphaPerC | Entered value: 0.00393
Input path: feeder.maximumConductorTemperatureC | Entered value: 20
Input path: feeder.minimumConductorTemperatureC | Entered value: 90

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_IEC60909_RADIAL_TRANSFORMER_ENVELOPE
Calculated output path: scope | Value: Single utility Thevenin source, one 2-winding transformer at rated reference tap, one LV feeder, radial balanced 3PH initial symmetrical maximum/minimum scenarios. Calculates nameplate transformer R/X, correction KT=0.95*cmax/(1+0.6*xT) and IEC-style voltage-factor current. Source equivalent short-circuit MVA is specifically defined as nominal V^2/|Zsource| (does NOT include a voltage factor). Not full IEC 60909-0:2026, not network generator/motor/IBR duty, tap-position rules or breaker interrupting qualification.
Calculated output path: highVoltageV | Value: 13800
Calculated output path: lowVoltageV | Value: 480
Calculated output path: transformer.ratedMVA | Value: 2.5
Calculated output path: transformer.shortCircuitVoltagePercent | Value: 5.75
Calculated output path: transformer.ratedLoadLossKW | Value: 18
Calculated output path: transformer.atRatedTapOnly | Value: true
Calculated output path: transformer.uncorrectedRohm | Value: 0.000663552
Calculated output path: transformer.uncorrectedXohm | Value: 0.005257491738775821
Calculated output path: transformer.relativeReactancePu | Value: 0.05704743640164736
Calculated output path: transformer.impedanceCorrectionKT | Value: 1.0104150471162217
Calculated output path: provenance.transformerReference | Value: ILLUSTRATIVE: replace with actual

approved 2500 kVA transformer OEM nameplate and loss test
Calculated output path: provenance.utilityReference | Value: ILLUSTRATIVE: replace with max/min POI equivalent utility study and matching X/R
Calculated output path: provenance.feederReference | Value: ILLUSTRATIVE LV cable sizing and installation; replace with approved conductor R20/X/route/temperature
Calculated output path: scenarios[0].scenario | Value: MAXIMUM
Calculated output path: scenarios[0].voltageFactor | Value: 1.1
Calculated output path: scenarios[0].conductorTemperatureC | Value: 20
Calculated output path: scenarios[0].utilityEquivalentMVA | Value: 350
Calculated output path: scenarios[0].utilityXOverR | Value: 10
Calculated output path: scenarios[0].sourceEquivalentLVImpedanceOhm.r | Value: 0.00006550187674982329
Calculated output path: scenarios[0].sourceEquivalentLVImpedanceOhm.x | Value: 0.0006550187674982329
Calculated output path: scenarios[0].correctedTransformerImpedanceOhm.r | Value: 0.0006704629253440631
Calculated output path: scenarios[0].correctedTransformerImpedanceOhm.x | Value: 0.005312248762948318
Calculated output path: scenarios[0].feederImpedanceOhm.r | Value: 0.004
Calculated output path: scenarios[0].feederImpedanceOhm.x | Value: 0.002
Calculated output path: scenarios[0].totalEquivalentImpedanceOhm.r | Value: 0.004735964802093887
Calculated output path: scenarios[0].totalEquivalentImpedanceOhm.x | Value: 0.007967267530446551
Calculated output path: scenarios[0].shortCircuitCurrent.realA | Value: 16805.624338409994
Calculated output path: scenarios[0].shortCircuitCurrent.imaginaryA | Value: -28271.938393861798
Calculated output path: scenarios[0].shortCircuitCurrent.magnitudeA | Value: 32889.68698467766
Calculated output path: scenarios[0].shortCircuitCurrent.angleDegrees | Value: -59.2715759691218
Calculated output path: scenarios[0].initialSymmetricalKA | Value: 32.88968698467766
Calculated output path: scenarios[0].screenPeakA | Value: 55105.1012606707
Calculated output path: scenarios[0].screenPeakKA | Value: 55.1051012606707
Calculated output path: scenarios[0].networkXOverR | Value: 1.6822902752410716
Calculated output path: scenarios[0].scope | Value: Initial symmetrical RMS and single radial passive-system exponential peak screening; no circuit-breaker interrupting or close-and-latch qualification
Calculated output path: scenarios[1].scenario | Value: MINIMUM
Calculated output path: scenarios[1].voltageFactor | Value: 0.95
Calculated output path: scenarios[1].conductorTemperatureC | Value: 90
Calculated output path: scenarios[1].utilityEquivalentMVA | Value: 220
Calculated output path: scenarios[1].utilityXOverR | Value: 8
Calculated output path: scenarios[1].sourceEquivalentLVImpedanceOhm.r | Value: 0.00012989819476978923
Calculated output path: scenarios[1].sourceEquivalentLVImpedanceOhm.x | Value: 0.0010391855581583138
Calculated output path: scenarios[1].correctedTransformerImpedanceOhm.r | Value: 0.0006704629253440631
Calculated output path: scenarios[1].correctedTransformerImpedanceOhm.x | Value: 0.005312248762948318
Calculated output path: scenarios[1].feederImpedanceOhm.r | Value: 0.005100400000000001
Calculated output path: scenarios[1].feederImpedanceOhm.x | Value: 0.002
Calculated output path: scenarios[1].totalEquivalentImpedanceOhm.r | Value: 0.005900761120113853
Calculated output path: scenarios[1].totalEquivalentImpedanceOhm.x | Value: 0.008351434321106632
Calculated output path: scenarios[1].shortCircuitCurrent.realA | Value: 14856.759460467198
Calculated output path: scenarios[1].shortCircuitCurrent.imaginaryA | Value: -21026.990981830735
Calculated output path: scenarios[1].shortCircuitCurrent.magnitudeA | Value: 25746.022050331823
Calculated output path: scenarios[1].shortCircuitCurrent.angleDegrees | Value: -54.75663077054178
Calculated output path: scenarios[1].initialSymmetricalKA | Value: 25.746022050331824
Calculated output path: scenarios[1].screenPeakA | Value: 41422.98388804437
Calculated output path: scenarios[1].screenPeakKA | Value: 41.42298388804437
Calculated output path: scenarios[1].networkXOverR | Value: 1.4153147621311426
Calculated output path: scenarios[1].scope | Value: Initial symmetrical RMS and single radial

passive-system exponential peak screening; no circuit-breaker interrupting or close-and-latch qualification
Calculated output path: maxInitialSymmetricalKA | Value: 32.88968698467766
Calculated output path: minInitialSymmetricalKA | Value: 25.746022050331824
Calculated output path: reference | Value: IEC 60909-0:2016 transformer K_T expression reproduced in Schneider Electric TP20180605 (2018); method remains an explicitly bounded screening, NOT full IEC 60909-0:2026 conformity
Calculated output path: limitations[0] | Value: Only radial single utility source and one two-winding transformer; no motor/generator/IBR/source contribution
Calculated output path: limitations[1] | Value: No LV transformer off-rated tap, load tap changer correction, unbalanced or ground faults
Calculated output path: limitations[2] | Value: Utility source input short-circuit MVA is a stated equivalent Z definition; it must not already include an IEC voltage factor
Calculated output path: limitations[3] | Value: LV cable resistance at temperature uses user-entered linear alpha; no skin/proximity derivation or sheath effects
Calculated output path: limitations[4] | Value: Exponential peak kappa only appropriate to simple passive equivalent R+jX source network
Calculated output path: limitations[5] | Value: No device protection trip-time, breaking or thermal withstand validation

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: transformer60909
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:33.629Z
Evidence / limitation: outputs.scope | Value: Single utility Thevenin source, one 2-winding transformer at rated reference tap, one LV feeder, radial balanced 3PH initial symmetrical maximum/minimum scenarios. Calculates nameplate transformer R/X, correction KT=0.95*cmax/(1+0.6*xT) and IEC-style voltage-factor current. Source equivalent short-circuit MVA is specifically defined as nominal V^2/|Zsource| (does NOT include a voltage factor). Not full IEC 60909-0:2026, not network generator/motor/IBR duty, tap-position rules or breaker interrupting qualification.
Evidence / limitation: outputs.provenance.transformerReference | Value: ILLUSTRATIVE: replace with actual approved 2500 kVA transformer OEM nameplate and loss test
Evidence / limitation: outputs.provenance.utilityReference | Value: ILLUSTRATIVE: replace with max/min POI equivalent utility study and matching X/R
Evidence / limitation: outputs.provenance.feederReference | Value: ILLUSTRATIVE LV cable sizing and installation; replace with approved conductor R20/X/route/temperature
Evidence / limitation: outputs.reference | Value: IEC 60909-0:2016 transformer K_T expression reproduced in Schneider Electric TP20180605 (2018); method remains an explicitly bounded screening, NOT full IEC 60909-0:2026 conformity
Evidence / limitation: outputs.limitations[0] | Value: Only radial single utility source and one two-winding transformer; no motor/generator/IBR/source contribution
Evidence / limitation: outputs.limitations[1] | Value: No LV transformer off-rated tap, load tap changer correction, unbalanced or ground faults
Evidence / limitation: outputs.limitations[2] | Value: Utility source input short-circuit MVA is a stated equivalent Z definition; it must not already include an IEC voltage factor
Evidence / limitation: outputs.limitations[3] | Value: LV cable resistance at temperature uses user-entered linear alpha; no skin/proximity derivation or sheath effects
Evidence / limitation: outputs.limitations[4] | Value: Exponential peak kappa only appropriate to simple passive equivalent R+jX source network
Evidence / limitation: outputs.limitations[5] | Value: No device protection trip-time, breaking or thermal withstand validation